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HW2

R5. What information is used by a process running on one host to identify a process running on another host?

- **IP address**: uniquely identifies the host
- Destination **Port Number**: Identifies the specific process/socket of the host.

R6. Suppose you wanted to do a transaction from a remote client to a server as fast as possible. Would you use UDP or TCP? Why?

- UDP because unlike TCP, UDP doesn't require a three-way handshake to send the data.

R12. Consider an e-commerce site that wants to keep a purchase record for each of its customers. Describe how this can be done with cookies.

- When a user visits the site, the website's server generates a unique ID number and returns it in a cookie header that is stored by the browser. From then on, the browser sends the ID back to the server identifying the user to the server containing all the information like name, credit card number, address, history, etc.

R13. How is it that QUIC can be connection-oriented even though it runs over UDP?

- The QUIC protocol manages things like connection, encryption, and data integrity within its own packet structure implemented at the application layer rather than the transport layer. QUIC only uses UDP for fast data delivery.

R16. Alice uses Gmail and Bob uses Yahoo! Mail. Suppose Alice sends an email message to Bob. List the series of application protocols that are used to get the message from Alice to Bob.

- Alice uses **SMTP** to push (send) the message to the Gmail server
  The Gmail server uses **DNS** to find where to the the mail
  The Gmail server uses **SMTP** to push the message to Yahoo! Mail server
  Bob uses **IMAP** or **HTTP** to pull (recieve) the message from Yahoo! Server on his device.

R18. What is the HOL blocking issue in HTTP/1.1? How does HTTP/2 attempt to solve it?

- In HTTP/1.1, HOL blocking happens when a large or slow request at the front of the line forces all subsequent requests to wait, stalling the connection. HTTP/2 fixes this by using **multiplexing**, which breaks data into independent frames that can be sent simultaneously over a single connection so one slow file doesn't block the rest.

R19. Is it possible for an organization's web server and mail server to have exactly the same alias for a hostname (for example, foo.com)? What would be the type for the RR that contains the hostname of the mail server?

- Yes, an organization can use the same hostname for both its web and mail servers. The Resource Record (RR) for the mail server would be an Mail Exchanged (MX) record.

P2 SMS, iMessage, Wechat, and WhatsApp are all smartphone real-time messaging systems. After doing some research on the Internet, for two of these systems (you can choose which ones) write one paragraph about the protocols they use. Then write a paragraph explaining how they differ. (This does not have to be overly detailed, just get a general idea of each!)

- WhatsApp uses XMPP to handle the message delivery system and the signal protocol for end to end encryption. IMessage also uses XMPP for messaging system along with APNs for delivery and PQ3 protocol for cryptography all of which ensure delivery, reliability, and security.

  The main difference is WhatsApp is cross-platform and works on any device with an internet connection while IMessage can only be used within Apple's ecosystem. Another difference is while Apple provides a much higher level of security through its use of security, its very limited relative to WhatsApp ability to handle massive group chats.

https://medium.com/@rajendra_51543/how-whatsapp-works-197bfc6d6b95